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Non-additive effects of different litter mixtures on soil carbon and nitrogen dynamics controlled by litter functional identity and diversity across two soils in the temperate forests of northeastern China

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Abstract

Background Understanding how plant litter diversity influences soil carbon (C) and nitrogen (N) dynamics is critical for explaining C and N cycling in forest ecosystems. However, the influence of litter functional traits on soil C and N dynamics across different forest soils remains unclear. In this study, we incubated 38 litter mixtures from 10 dominant tree species with contrasting functional traits in soils from natural secondary forests (NSF) and *Larix olgensis* plantations (LOP) in northeastern China. We measured C and N mineralization, dissolved organic carbon (DOC), and microbial biomass carbon (MBC) to quantify non-additive effects. In addition, we evaluated the effects of litter functional identity (Community-weighted mean, CWM) and functional diversity (Rao's *Q*) on non-additive effects across different soils.

Results The non-additive effects of mixed leaf litter on soil C and N dynamics varied across the two soils. The proportion of non-additive effects on C and N mineralization, DOC, and MBC ranged from 17.8% to 60.7% in both soils. Non-additive effects were generally more frequent in NSF soils than in LOP soils, which may be attributed to the higher organic matter content and more diverse microbial communities. Trait-based models showed that functional diversity was the dominant predictor of non-additive effects in NSF soils, explaining 89.3% and 84.3% of the variations in C and N mineralization, respectively. However, in LOP soils, trait effects were more function-specific; functional diversity governed C mineralization, whereas functional identity had a greater influence on N mineralization. Specific traits, such as Rao's *Q* of TN and TC, and CWM of TC, emerged as key regulators of non-additive effects.

Conclusions Our findings underscore the importance of integrating trait-based frameworks with different forest soils to understand the non-additive effects of litter mixtures. By identifying how specific traits mediate soil C and N dynamics differently across the two soils, this study provides a theoretical basis for predicting biogeochemical responses to litter diversity, with implications for forest biodiversity conservation and soil fertility improvement strategies.

Keywords Litter diversity, Forest soil, Non-additive effect, Functional identity, Functional diversity, Carbon and nitrogen mineralization

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Introduction

Plant diversity plays a fundamental role in determining ecosystem functions and influences key ecological processes such as nutrient dynamics, primary productivity and decomposition (Schuldt et al. 2018; Chen et al. 2022, 2023; Zheng et al. 2024). Litter decomposition is a key process in forest ecosystems that links plants and soil by cycling essential elements such as carbon (C) and nitrogen (N) (Mori et al. 2020; Zhang et al. 2023). As litter decomposes, it releases nutrients that substantially enhance soil fertility and influence microbial communities (Cotrufo et al. 2015; Shah et al. 2024). The key factor regulating the rate of litter decomposition is its chemical composition, which varies widely between plant species (Cornwell et al. 2008). Mixing litter from different species can cause non-additive effects that differ from the expected sum of single-species contributions (Liu et al. 2020; Xiao et al. 2020). These interactions may be synergistic, accelerating decomposition and nutrient release, or antagonistic, inhibiting these processes (Kou et al. 2020; Porre et al. 2020). Such non-additive effects are not limited to litter decomposition, but are also observed in soil C and N mineralization and nutrient availability, where the nutrient release dynamics similarly diverge from expectations based on single-species litter inputs (Meier and Bowman 2008; Chen et al. 2020).

Uncovering the mechanisms underlying these non-additive effects is central to advancing our understanding of forest biodiversity conservation and improving soil fertility (van der Plas et al. 2016). In recent years, trait-based ecology has offered new insights into how the functional traits of litter influence soil processes, emphasizing the importance of litter functional trait diversity in shaping soil functions (Man et al. 2020; Buchanan et al. 2024; Edwards et al. 2025). Functional traits, which are morphological, physiological, or phenological characteristics that affect organism performance or ecosystem processes (Violle et al. 2007; Pérez-Harguindeguy et al. 2013), serve as mechanistic links between species composition and ecosystem functioning. The diversity of litter functional traits, encompassing both functional identity (e.g., community-weighted mean values) and functional diversity (e.g., trait variability or dissimilarity among species), plays a pivotal role in determining the direction and magnitude of soil responses to litter mixtures. For instance, high functional complementarity among species can enhance resource-use efficiency and microbial niche differentiation, thereby promoting synergistic non-additive effects (Kou et al. 2020), whereas a strong dominance of recalcitrant litter components may generate antagonistic effects (Liu et al. 2020). Therefore, a trait-based perspective provides a useful foundation for examining how litter mixtures influence belowground processes.

Recent studies have increasingly focused on the effects of mixtures of litter species on belowground processes, revealing that non-additive effects are common during decomposition. For example, Meier and Bowman (2008) reported that the composition and diversity of litter chemical traits, rather than species richness, were the strongest predictors of soil C and N cycling. Similarly, Chen et al. (2015) reported that the non-additive effects of litter diversity significantly influenced greenhouse gas emissions (CO_2, CH_4, and N_2O) in alpine steppe soils, with the direction of these effects being closely linked to the specific chemical properties of the litter. Other studies have demonstrated that the chemical properties of litter mixtures, such as N content and C/N ratio, drive microbial activity, mineralization rates, and soil nutrient dynamics, often more than species richness alone (Jiang et al. 2013; Chen et al. 2017, 2018; Mao et al. 2017). Despite these findings, the relative importance of functional identity versus functional diversity in shaping these non-additive effects remains poorly understood, particularly within forest systems.

Temperate forest ecosystems support high biodiversity and include a mixture of deciduous and coniferous tree species, leading to considerable variations in litter quality and decomposition dynamics (Yuan et al. 2021; Yang et al. 2022). The interaction between leaf litter and soil regulates nutrient cycling in these systems (Yang and Zhu 2015). Moreover, soils in temperate forests are highly heterogeneous—ranging from the nutrient-poor soils of coniferous forests to the nutrient-rich soils of deciduous forests, further complicating the outcomes of litter decomposition and its influence on belowground biogeochemical processes (Yang and Zhu 2015; Meng et al. 2025). However, background soil conditions are often overlooked in experimental research (Prosser and Martiny 2020). Incubation experiments are commonly conducted to control confounding variables under standardized conditions; however, homogeneous soil conditions are often used. This simplification may obscure critical context-dependent interactions between litter traits and soil processes, thereby constraining the generalizability of the experimental results to natural forest ecosystems. Soil properties such as pH, organic matter content, and microbial community structure vary across forest types. Therefore, it is crucial to examine how soil conditions influence the effects of litter mixtures on belowground biogeochemical processes.

To address these gaps, in this study, we conducted laboratory incubations using soil from two ecologically distinct forest types in northeastern China: natural secondary forest (NSF) and *Larix olgensis* plantation (LOP) soil. Soil samples were collected from four spatially independent plots to better reflect natural variability. A series

of single- and mixed-species leaf litter treatments were applied to each soil sample to assess their effects on soil C and N dynamics. Specifically, we addressed the following questions: (1) Do litter mixtures induce non-additive effects on soil C and N dynamics that vary across different soil conditions? (2) Which aspects of litter functional traits most strongly predict non-additive effects? (3) Do different forest soil conditions mediate the strength or direction of litter functional traits with non-additive effects? By integrating trait-based theory, non-additive effects, and soil conditions, our study provides insights into how litter functional diversity and identity jointly influence belowground biogeochemical processes. These findings advance our mechanistic understanding of biodiversity–ecosystem functioning relationships and have implications for optimizing litter inputs to enhance soil fertility and promote sustainable forest management.

Materials and methods

Study site and sampling

Leaf litter and soils were collected from the Qingyuan Forest CERN (Chinese Ecosystem Research Network), Chinese Academy of Sciences, situated in Liaoning Province, Northeast China (41°51′ N, 124°54′ E). The climate is classified as continental monsoon, characterized by warm, humid summers and cold, dry winters with a mean annual temperature of 4.7 °C. Annual precipitation ranges from 700 to 850 mm, over 80% of which falls between June and August. The frost-free period extends from 120 to 139 days annually. The soils in the study area are typical brown forest soils. Vegetation in the study area is primarily composed of natural secondary forests, with dominant species including *Acer mono*, *Fraxinus rhynchophylla*, *Juglans mandshurica*, *Quercus mongolica* and other tree species. Owing to the increasing demand for timber, extensive areas of secondary forests have been converted into *Larix olgensis* plantations, resulting in a mosaic of plantation and secondary forest landscapes (Yang et al. 2022).

In November 2021, senescent but intact leaves were collected from the forest floor within the study area. Leaf litter was collected from 10 locally dominant tree species, including *A. mono*, *Q. mongolica*, *J. mandshurica*, *Carpinus cordata*, *F. rhynchophylla*, *Fraxinus mandshurica*, *Sorbus alnifolia*, *Populus davidiana*, *Pinus koraiensis*, and *L. olgensis*. The litter was oven-dried at 60 °C to a constant weight and subsequently cut into uniform 1 × 1-cm segments. The prepared litter was then placed in nylon mesh bags (1-mm mesh size) for subsequent experiments.

In May 2022, soil samples were collected from four spatially distinct paired sites within the study area, comprising natural secondary forests and *L. olgensis* plantations. These sites are long-term ecological monitoring plots; detailed site information is described in Yang and Zhu (2015). At each site, a 30 × 30-m plot was established, and 15 random sampling points were selected within each plot. Soils from the top 0–20 cm of the mineral horizon were collected after removing the surface litter. The soil samples from the 15 points within each plot were then combined to form a mixed sample to ensure consistency in soil properties. This procedure was applied to both the secondary forest and *L. olgensis* plantation plots, yielding four homogeneous soil samples each from the secondary forests and the *L. olgensis* plantations. All soil samples were collected using a soil auger, sieved through a 2-mm mesh to remove stones, roots, and other debris, and thoroughly homogenized for each site. Four replicates per forest soil were kept separate to maintain spatial independence for subsequent analyses. The initial soil properties were as follows: for NSF soils, total C, total N, total P, mineral N, and available P: 50.8 g kg^{-1}, 4.4 g kg^{-1}, 1.0 g kg^{-1}, 38.9 mg kg^{-1} and 2.6 mg kg^{-1}, respectively. For LOP soils, total C, total N, total P, mineral N, and available P were 31.3 g kg^{-1}, 2.8 g kg^{-1}, 1.7 g kg^{-1}, 19.5 mg kg^{-1} and 4.1 mg kg^{-1}, respectively (Table S1).

Experiment design

A microcosm incubation experiment was conducted to investigate the effects of plant litter mixing on soil C and N dynamics in different soils. To minimize the initial respiration flush typically induced by soil disturbance and rewetting, the prepared soil mixtures were pre-incubated for 15 days at 25 °C and adjusted to 60% of their water-holding capacity before the experiment began. Following pre-incubation, each microcosm consisted of moist field soil (adjusted to 60% water-holding capacity) in an amount equivalent to 50 g of dry weight, combined with 1 g of oven-dried leaf litter, yielding a litter-to-soil dry mass ratio of 1:50. The experimental design included 38 distinct litter treatments: 10 single-species treatments, 22 two-species mixtures, 5 four-species mixtures, and 1 eight-species mixture. The mixtures were designed primarily along a gradient of litter N concentrations and included a wide range of functional diversity to capture strong functional differences. All mixtures were prepared by combining component species with equal dry masses (see Tables S2 and S3 for details). Each treatment was applied to soil from two forest types: NSF and LOP. Soil was collected from four spatially independent plots within each forest type and treated as separate replicates. Four spatial replicates were established for each litter–soil combination. Each replicate was further divided into two temporal subsamples to allow destructive sampling at different time points. Additionally, control microcosms (i.e., with no litter added) were established for both forest

soils, following the same replication structure. The experiment consisted of 624 microcosms, derived from 39 litter treatments (38 litter treatments plus one with no litter added) × 2 different forest soils × 2 incubation periods × 4 replicates. Leaf litter bags were then placed on the soil surface. Each litter–soil combination was incubated in a 500-mL polyethylene jar, sealed, and maintained in the dark at 25 °C. The soil moisture was adjusted to 60% water-holding capacity and maintained throughout the 180-day incubation period by weighing each jar every 3–4 days and adding sterile distilled water as needed to adjust the mass (Yang and Zhu 2015).

Cumulative CO_2 release, used to assess carbon mineralization, was measured at 10 time points: 3, 7, 15, 30, 45, 60, 90, 120, 150, and 180 days. A static alkali absorption method was employed, in which a 10-mL glass vial containing 5 mL of 0.5 mol L^{-1} NaOH solution was inserted into each microcosm with the vial opening positioned slightly above the soil surface to trap any CO_2 emitted via soil respiration. The volume and concentration of NaOH used were sufficient to absorb all of the CO_2 produced during each interval. At each interval, the vial was replaced with a fresh one, and the absorbed CO_2 was quantified by acid–base titration. Destructive soil sampling was conducted at 90 and 180 days to assess chemical and microbial properties. Inorganic nitrogen (NH_4^+ and NO_3^-) was measured at both time points. Dissolved organic carbon (DOC) and microbial biomass carbon (MBC) were analyzed on day 180.

Litter and soil characteristics analyses

The chemical traits of litter were comprehensively analyzed before and after 180 days of decomposition. The leaf litter was oven-dried, ground using a planetary ball-mill, and passed through a 100-mesh sieve. The total C and N contents were determined using an elemental analyzer (Elementar Analysensysteme GmbH, Germany). The total phosphorus (P) content was measured using the molybdenum antimony colorimetric method following sulfuric acid digestion (Olsen and Sommers 1982). The total potassium (K), calcium (Ca), iron (Fe), magnesium (Mg), manganese (Mn), zinc (Zn), and copper (Cu) concentrations were determined by inductively coupled plasma–optical emission spectrometry (ICP-OES; Agilent 5100, Australia) following HNO_3–$HClO_4$ digestion (Melaku et al. 2003).

The litter lignin, cellulose, and hemicellulose contents were determined following the method described by Van Soest et al. (1991) using an ANKOM A2000i Fiber Analyzer (ANKOM Technology, USA). The procedure involved sequential analysis of neutral detergent fiber (NDF), acid detergent fiber (ADF), and acid detergent lignin (ADL). The hemicellulose, cellulose, and lignin contents were calculated as NDF minus ADF, ADF minus ADL, and ADL, respectively. All fiber measurements were performed in triplicate, with a coefficient of variation below 2%, and standard reference materials were used for quality assurance.

Soil C mineralization was quantified using the static alkali absorption method. During incubation, a 10-mL glass vial containing 5 mL of 0.5 mol L^{-1} NaOH was added to each microcosm to capture CO_2 released via soil respiration. The NaOH vial was replaced at designated time points, and the residual NaOH was titrated with standardized 0.1 mol L^{-1} HCl. The amount of CO_2 released was calculated from the amount of NaOH consumed using stoichiometric relationships based on the CO_2–NaOH reaction equations. The titrant concentration was pre-verified using the boric acid–HCl titration method, and CO_2 release was normalized to soil dry mass (μg CO_2 g^{-1} soil; Yang and Zhu 2015).

Soil inorganic N, including ammonium (NH_4^+) and nitrate (NO_3^-), was measured in fresh soil samples collected at 90 and 180 days. For each sample, 5 g of soil was extracted with 25 mL of 2 mol L^{-1} KCl by shaking for 30 min, followed by filtration. The NH_4^+ and NO_3^- concentrations in the extract were determined colorimetrically using a flow injection analyzer.

DOC was analyzed using the K_2SO_4 extraction method. Specifically, 5 g of soil was shaken with 20 mL of 0.5 mol L^{-1} K_2SO_4 for 30 min at 150 rpm and filtered. The extract was diluted and analyzed for organic carbon content using a Multi N/C 3000 analyzer (Elementar Analysensysteme GmbH, Germany). MBC was determined using the chloroform fumigation–extraction method (Vance et al. 1987). For each analysis, one set of samples was fumigated with ethanol-free chloroform for 24 h in a vacuum desiccator, and the other was used as the unfumigated control. Both were extracted using K_2SO_4 as described above. Extractable C was analyzed using the same TOC analyzer, and MBC was calculated as the difference between the fumigated and unfumigated samples divided by correction factors of 0.45 (Vance et al. 1987).

Data analyses

All statistical analyses were conducted using R software (version 4.2.2). Prior to statistical analyses, the residuals of all variables were tested for normality using the Shapiro–Wilk and Kolmogorov–Smirnov test. The results indicated that the data were normally distributed. To evaluate the variation in litter chemical properties among the 10 tree species before incubation and after 180 days of decomposition, one-way analysis of variance (ANOVA) was performed separately for each trait. Similarly, ANOVA was applied to assess how

individual litter species affected soil response variables, including cumulative CO_2 release and the inorganic N, DOC, and MBC concentrations, during the decomposition process.

To determine whether litter mixtures exerted additive or non-additive effects on soil function, we estimated the expected values for each two-, four- and eight-species combination based on the average response of the corresponding single-species treatments. The expected values for each litter mixture were calculated as follows (Meier and Bowman 2010):

$$B_{litter} = \sum_{i=1}^{S} \frac{R_i}{S}$$

where R_i denotes the soil response when species i is present alone and S represents the total number of species in the litter mixtures. These expected values were compared with the observed values from the mixtures using paired t-tests. A significant difference ($P < 0.05$) between the observed and expected values was interpreted as evidence of a non-additive interaction. The magnitude and direction of the litter-mixture effect were further quantified as the relative difference between the observed and expected values, calculated as follows (Wardle et al. 1997):

$$\text{Litter mixture effect} = (O - E)/E$$

where O represents the observed value and E is the expected value. Positive values indicate synergistic effects, and negative values indicate antagonistic effects. To investigate the role of litter trait diversity in shaping non-additive decomposition outcomes, we quantified a comprehensive set of 32 functional trait metrics for each litter mixture, including 16 community-weighted means (CWMs) and 16 Rao's quadratic entropy (Rao's Q) measures, based on initial trait values (Fig. S1). The CWM represents functional identity by averaging trait values across species weighted by their relative contribution to the litter mixture, thus reflecting the dominant functional characteristics of the mixture. Rao's Q captures functional diversity by quantifying the trait dissimilarity among species within mixtures, thereby reflecting the extent of complementarity or variability in trait values. For each trait, the CWM and Rao's Q were calculated as follows (Garnier et al. 2004; Botta-Dukat 2005):

$$\text{CWM} = \sum_{i=1}^{n} p_i \times \text{trait}_i$$

where p_I represents the proportion of species i in the mixture, and trait_I is its corresponding trait value.

$$\text{Rao}_i = \sum_{i=1}^{n} \sum_{j=1}^{n} p_i p_j \times d_{ij}$$

where p_i and p_j denote the relative abundances of species i and j in the mixture, and d_{ij} represents the Euclidean-based trait distance between them in multidimensional trait space. All predictor variables, and the response variable, were standardized prior to analysis to facilitate interpretation and allow for comparability of effect sizes across variables.

We applied a two-step modeling procedure to identify the key functional traits that contributed to the observed patterns. First, a full linear model including all 32-trait metrics as fixed effects was constructed using the lm function in R. To reduce model complexity and eliminate redundant predictors, we performed stepwise model selection based on Akaike's Information Criterion (AIC), using the stepAIC function from the MASS package. Following variable selection, we assessed the multicollinearity among the retained predictors by calculating the variance inflation factors (VIFs) using the vif function from the car package. Any variable with a VIF exceeding five was excluded to ensure statistical reliability and prevent inflated standard errors (O'Brien 2007). A refined linear model was then re-fitted using only a subset of predictors with acceptable collinearity.

The set of trait predictors was subsequently used to fit a linear mixed-effects model using the lmer function from the lme4 package. In this model, trait variables were specified as fixed effects, whereas different soil replicates were included as random effects to account for the spatial heterogeneity across sampling locations. To evaluate the explanatory power of the model, we computed the marginal and conditional R^2 values, which represented the variance explained by fixed effects alone and by both fixed and random effects, respectively (Nakagawa and Schielzeth 2012). These metrics were calculated using the r.squaredGLMM function from the MuMIn package. Given the absence of significant differences in DOC across treatments, as shown in Fig. 2a, and the limited non-additive effects observed for MBC, with non-additive effects of 32.1% and 17.8% in NSF and LOP soils, respectively, these two variables were excluded from the modeling analysis.

Results

Chemical traits of single-species leaf litters before and after 180 days of decomposition

All initial leaf litter chemical traits differed significantly among the 10 tree species ($P < 0.05$; Table 1). Notably, *C. cordata* had the highest N concentration (15.1 g kg^{-1}), twice that of *P. koraiensis*, which also had the highest

Table 1 Initial chemical characteristics of leaf litter among tree species

Tree species	C (g kg^{-1})	N (g kg^{-1})	P (g kg^{-1})	C/N	C/P	N/P	K (mg kg^{-1})	Ca (mg kg^{-1})	Mg (mg kg^{-1})	Fe (mg kg^{-1})	Mn (mg kg^{-1})	Zn (μg kg^{-1})	Cu (μg kg^{-1})	Lignin (%)	Cellulose (%)	Hemicellulose (%)
Acer mono	**452.8** 0.3f	**14.0** 0.1c	**1.4** 0.1b	**32.3** 0.1g	**326.4** 13.6bc	**10.1** 0.4cd	**3.3** 0.5d	**23.3** 3.2c	**4.0** 1.8a	**0.6** 0.1a	**0.3** 0.0c	**21.5** 1.7cd	**8.5** 3.2b	**18.4** 1.1b	**17.0** 0.9b	**43.6** 0.9a
Quercus mongolica	**471.4** 0.4d	**9.6** 0.7g	**1.3** 0.1b	**55.7** 0.2b	**363.6** 15.3b	**6.5** 0.3f	**4.0** 0.1c	**13.3** 0.4d	**1.9** 0.1de	**0.3** 0.0d	**0.5** 0.0a	**19.0** 2.4d	**5.5** 1.8c	**12.0** 0.1e	**14.3** 0.3e	**12.1** 0.2e
Juglans mandshurica	**484.3** 0.2c	**11.6** 0.0e	**1.5** 0.0b	**41.7** 0.1d	**335.1** 10.1bc	**8.0** 0.3e	**3.6** 0.2d	**29.5** 1.5a	**2.1** 0.1cd	**0.2** 0.0e	**0.2** 0.0c	**26** 4.7bcd	**11.1** 2.8a	**13.8** 0.1d	**11.1** 0.0g	**8.6** 0.1f
Carpinus cordata	**452.8** 0.2f	**15.1** 0.0a	**1.9** 0.0a	**29.9** 0.0i	**245.1** 2.2d	**8.2** 0.1e	**6.0** 0.2a	**21.6** 0.5c	**2.8** 0.1bc	**0.3** 0.1cd	**0.2** 0.0c	**20.8** 0.3cd	**7.5** 0.5bc	**9.8** 0.2f	**16.3** 0.3c	**20.9** 0.5b
Fraxinus rhynchophylla	**467.6** 5.4de	**14.1** 0.3c	**1.1** 0.0c	**33.2** 0.3f	**428.7** 6.8a	**12.9** 0.2a	**3.6** 0.1d	**30.0** 0.8a	**3.3** 0.5ab	**0.2** 0.1e	**0.1** 0.0d	**30.9** 13.6bc	**8.9** 0.6ab	**8.8** 0.2g	**14.9** 0.5de	**13.0** 0.8d
Fraxinus mandshurica	**462.0** 0.2e	**14.6** 0.0b	**1.3** 0.1b	**31.6** 0.1h	**353.5** 19.7bc	**11.2** 0.6bc	**2.4** 0.1e	**26.8** 1.5b	**2.8** 0.4bc	**0.4** 0.0c	**0.1** 0.0de	**50.2** 13.1a	**8.8** 2.0ab	**12.8** 0.3e	**12.2** 0.6f	**14.7** 0.4c
Sorbus alnifolia	**466.4** 0.4de	**10.4** 0.0f	**1.8** 0.1a	**44.8** 0.2c	**263.9** 10.0d	**5.9** 0.2f	**6.1** 0.5a	**22.1** 1.9c	**2.8** 0.2bc	**0.5** 0.1b	**0.1** 0.0de	**22.0** 2.4cd	**7.1** 1.5bc	**21.1** 0.5a	**15.3** 0.6d	**11.9** 0.4e
Populus davidiana	**492.3** 1.0b	**14.5** 0.1b	**1.7** 0.1a	**33.9** 0.2f	**300.4** 19.2cd	**8.9** 0.6de	**4.4** 0.1b	**22.0** 0.6c	**2.2** 0.1cd	**0.3** 0.0d	**0.4** 0.0b	**57.3** 10.5a	**7.5** 0.9bc	**16.3** 0.5c	**14.7** 0.5de	**13.4** 0.4d
Pinus koraiensis	**501.3** 0.2a	**67.7** 0.0 h	**1.1** 0.1c	**80.3** 0.5a	**474.3** 35.7a	**5.9** 0.5f	**0.3** 0.0g	**6.4** 0e	**1.0** 0.0ef	**0.2** 0.0e	**0.6** 0.1a	**20.7** 0.6cd	**7.3** 0.3bc	**18.3** 0.7b	**21.8** 0.3a	**13.1** 1.2d
Larix olgensis	**488.0** 3.0bc	**12.2** 0.1d	**1.1** 0.1c	**40.1** 0.1e	**464.3** 32.6a	**11.6** 0.8b	**1.9** 0.1f	**2.8** 0.1f	**0.7** 0.0f	**0.1** 0.0e	**0.1** 0.0e	**32.7** 1.6b	**6.0** 0.4c	**15.6** 0.8c	**17.4** 0.4b	**11.5** 0.2e

The lowercase letters in this table represent the results of multiple comparisons. Different letters indicate significant differences between groups, while the same letter indicates no significant difference between groups

C/N ratio. Additionally, the N/P ratio was the highest in *F. rhynchophylla* and relatively low in *Q. mongolica*, *S. alnifolia,* and *P. koraiensis*. The hemicellulose concentration varied substantially, with the highest value in *A. mono* and lowest in *J. mandshurica*. Furthermore, the cellulose concentration was the lowest in *J. mandshurica* (11.1%) and highest in *P. koraiensis* (21.8%). The lignin concentration peaked in *S. alnifolia* (21.1%) and was the lowest in *F. mandshurica* (8.8%). Among the nutrients, *S. alnifolia* had the highest K concentration. Both *F. rhynchophylla* and *J. mandshurica* showed relatively higher Ca concentrations than the other litter types. *A. mono* exhibited the highest concentrations of Mg and Fe across all species. In addition, *F. rhynchophylla* had a high Zn concentration, whereas *J. mandshurica* had a higher concentration of Cu.

After 180 d of decomposition, the litters of *F. mandshurica* and *F. rhynchophylla* exhibited the greatest mass loss, approaching 50%, which was significantly higher than that of other species ($P < 0.05$; Table S4). In contrast, *P. koraiensis* and *L. olgensis* showed significantly lower mass loss. There were significant differences in all chemical traits of the leaf litter between the 10 tree species. Following decomposition, the C and P contents of the different litter types decreased, whereas the N content increased.

Effects of single litter-type decomposition on soil C and N dynamics

Cumulative C mineralization differed significantly between tree species and soils from the two forest types ($P < 0.05$; Fig. 1a, c). Specifically, in both NSF and LOP soils, *F. mandshurica* and *A. mono* litter exhibited relatively high CO_2 release. In addition, significant interspecific variations in N mineralization were observed between the two soils (Fig. 1b, d). In the NSF soils, the highest N mineralization rates were associated with *F. mandshurica*. In the LOP soils, *F. mandshurica* maintained relatively high N mineralization rates after both 90 and 180 days.

DOC and MBC varied significantly between the two soils (Fig. 2a, b). In the NSF soils, the DOC concentrations were relatively high and showed limited interspecific variation, with no significant differences among the species. MBC exhibited more pronounced species-level differences in NSF soils than in LOP soils. In the NSF soil, litter from *F. rhynchophylla* had the highest MBC, whereas *P. koraiensis* and *L. olgensis* had the lowest MBC. In contrast, no significant interspecific differences were detected in the LOP soils.

Effects of leaf litter diversity on soil C and N dynamics across the two soils

The effect of different litter mixtures on soil C and N dynamics varied significantly between the two soils ($P < 0.05$; Figs. 3, 4). Notably, C mineralization, N mineralization, DOC, and MBC exhibited distinct patterns between NSF and LOP soils. In NSF soils, the effects on C mineralization were predominantly non-additive (60.7%), with synergistic effects accounting for the majority (53.6%). Only two litter combinations, *L. olgensis* + *C. cordata* (Lg + Cc) and *J. mandshurica* + *C. cordata* (Jm + Cc), exerted antagonistic effects. Conversely, in LOP soils, non-additive effects accounted for 35.7%, with synergistic and antagonistic effects constituting 25.0% and 10.7%, respectively. NSF soils exhibited more pronounced synergistic interactions between the different litter mixtures, whereas LOP soils showed a combination of synergistic and antagonistic effects, with a weaker overall non-additive effect. For N mineralization, NSF soils exhibited non-additive, antagonistic, and synergistic effects of 46.4%, 28.6%, and 17.8%, respectively. In LOP soils, the proportion of significant non-additive effects was lower, and antagonistic and synergistic effects each accounted for 50% of the cases.

Regarding DOC, the non-additive effect in NSF soils was 42.9%, with synergistic effects contributing 28.6%. In LOP soils, the non-additive effect was 32.1%, with antagonistic effects accounting for 21.4%. For MBC, nine significant non-additive effects were observed in the NSF soils, all of which were antagonistic. In LOP soils, only five significant non-additive effects were observed.

Divergent contributions of functional identity and diversity to non-additive effects on C and N mineralization across the two soils

Litter functional traits explained a substantial proportion of the variations in non-additive C mineralization across the two soils (Table 2). However, the relative contributions of functional identity (CWM) and functional diversity (Rao's *Q*) differed considerably. In NSF soils, functional diversity metrics dominated, accounting for 89.3% of the explained variance, whereas functional identity contributed only 10.7%. Specifically, Rao's *Q* of TC had a negative effect on non-additive C mineralization ($P < 0.001$), whereas Rao's *Q* of TN had a positive effect ($P < 0.001$). Among the CWM traits, both the CWM of TC ($P < 0.05$) and cellulose ($P < 0.01$) were positively associated with non-additive C mineralization, although their overall contribution was limited. In contrast, LOP soils showed a more balanced contribution between trait identity (35.6%) and functional diversity (64.4%) relative to NSF soils. The CWM of TC and hemicellulose was positively associated with non-additive C mineralization

Table 2 Summary of the best-fit linear mixed-effects models explaining soil nutrient dynamics in NSF and LOP soils based on community functional composition (CWM) and functional diversity (Rao's *Q*)

NSF soil					LOP soil				
Fixed terms	**Estimate**	**Std. Error**	***P***	**Variance explained (%)**	**Fixed terms**	**Estimate**	**Std. Error**	***P***	**Variance explained (%)**
A. Soil carbon mineralization									
TC CWM	0.28	0.11	< 0.05	10.69	TC CWM	0.82	0.19	< 0.001	35.60
Cellulose CWM	0.39	0.14	< 0.01		Hemicellulose CWM	0.62	0.19	< 0.001	
TC Rao's *Q*	-0.87	0.15	< 0.001	89.31	TC Rao's *Q*	− 0.74	0.16	< 0.001	64.40
TN Rao's *Q*	0.57	0.1	< 0.001						
Marginal $R^2 = 0.52$					Marginal $R^2 = 0.51$				
B. Soil nitrogen mineralization									
N/P CWM	0.27	0.17	0.1	15.69	TC CWM	− 0.5	0.15	< 0.01	60.32
TC Rao's *Q*	0.27	0.13	0.05	84.31	TN CWM	− 0.48	0.15	< 0.01	
TN Rao's *Q*	0.37	0.16	< 0.05		TN Rao's *Q*	0.48	0.15	< 0.01	39.68
Marginal $R^2 = 0.18$					Marginal $R^2 = 0.43$				

Marginal R^2 indicates the variance explained by fixed effects. $P < 0.05$ was considered statistically significant

($P < 0.001$). Rao's *Q* of TC showed a significant negative effect ($P < 0.001$).

In contrast to non-additive C mineralization, the trait-based drivers of non-additive nitrogen mineralization varied more substantially between the soils from the two forest types. In NSF soils, functional trait predictors explained a moderate proportions of the variation in soil non-additive N mineralization, with functional diversity (Rao's *Q*) explaining the majority (84.3%) and functional identity (CWM) contributing only 15.7%. Rao's Q of TN was positively associated with non-additive N mineralization ($P < 0.05$). In contrast, a greater proportion of the variation was explained in LOP soils, with trait identity (60.3%) making a stronger contribution than functional diversity (39.7%). Regarding CWM, the values for both TC and TN were negatively associated with non-additive N mineralization ($P < 0.01$). Conversely, Rao's *Q* of TN exhibited a significant positive effect ($P < 0.01$).

Discussion

Species-specific litter traits drive variations in soil C and N dynamics

Interspecific variation in initial litter chemistry, particularly the N concentration, C/N ratio, and lignin content, led to different effects on soil C and N dynamics during decomposition. This aligns with previous findings that litter with higher N and lower lignin contents decomposes more rapidly, leading to greater microbial activity and nutrient release (Cornwell et al. 2008). For example, *F. mandshurica* and *F. rhynchophylla*, which have relatively high N concentrations and low lignin contents, decompose more rapidly and stimulate higher CO_2 release and N mineralization. In contrast, coniferous species such as *P. koraiensis* and *L. olgensis*, characterized by lower N and higher concentrations of structural compounds (e.g., cellulose and lignin), exhibited slower decomposition and weaker nutrient release, which is consistent with previous reports of coniferous litter resisting microbial degradation because of its recalcitrant chemical structure (Yang et al. 2022), as well as its influence on specific fungal communities and enzyme activities (Zhang et al. 2019). Moreover, some broadleaf species, such as *J. mandshurica*, produce antibacterial compounds that may further suppress microbial activity and influence decomposition processes (Mortier et al. 2020).

Effects of leaf litter diversity on soil C and N dynamics across the two soils

Our results demonstrated that the non-additive effects of mixed leaf litter on soil C and N dynamics are context-dependent. The proportion of non-additive effects across functions such as C mineralization, N mineralization, DOC, and MBC ranged from 17.8% to 60.7%, relatively lower than values reported in previous studies, which often exceeded 80% for parameters such as soil CO_2 emissions, inorganic N, and MBC (Chen et al. 2017). Such differences likely reflect variations in soil conditions and experimental contexts. In this study, we used spatially independent soil samples from two contrasting forest soils with different C and nutrient contents (Yang and Zhu 2015). This design, intended to reflect more field-relevant conditions, may have introduced greater

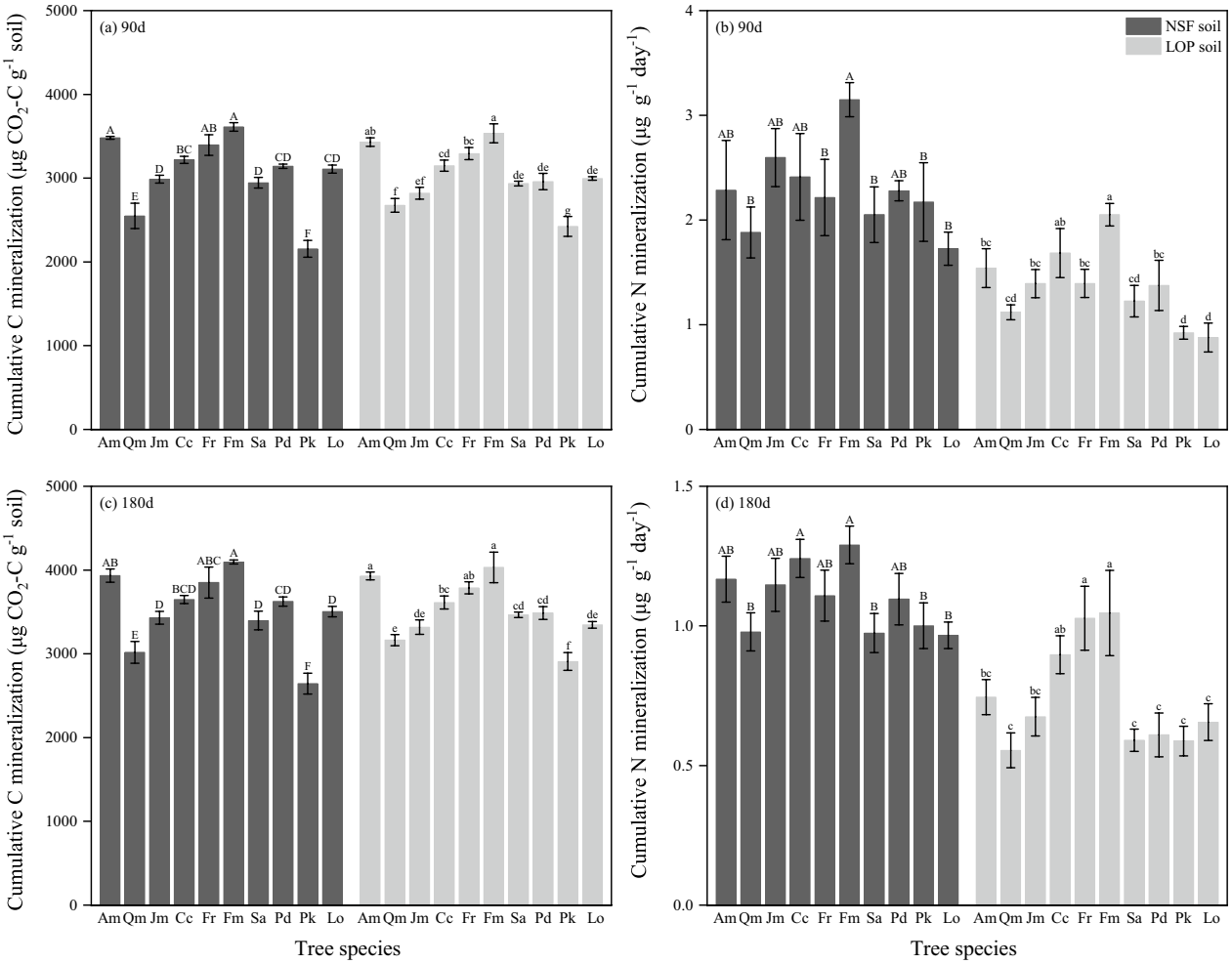


Fig. 1 Effects of different single-species litters on soil C and N dynamics in natural secondary forests (NSF) and *Larix olgensis* plantations (LOP) soils at 90 days (**a**–**b**) and 180 days (**c**–**d**). (**a**, **c**) Cumulative carbon (C) mineralization (μg CO_2-C g^{-1} soil); (**b**, **d**) Cumulative nitrogen (N) mineralization (μg g^{-1} day^{-1}). Black bars represent NSF soil, and gray bars represent LOP soil. Different lowercase and uppercase letters above bars indicate significant differences among litter treatments within each soil conditions ($P < 0.05$). Tree species abbreviations: Am – *Acer mono*, Qm – *Quercus mongolica*, Jm – *Juglans mandshurica*, Cc – *Carpinus cordata*, Fr – *Fraxinus rhynchophylla*, Fm – *Fraxinus mandshurica*, Sa – *Sorbus alnifolia*, Pd – *Populus davidiana*, Pk – *Pinus koraiensis*, Lo – *Larix olgensis*

background variability, thereby reducing the consistency and magnitude of the observed non-additive effects. Our findings suggest that, under field-relevant, heterogeneous conditions, the strength of litter-driven non-additive effects may be weaker than that in controlled or homogenized systems. Moreover, the observed variability in fertility conditions underscores the modulating role of fine-scale soil heterogeneity on litter–soil interactions, emphasizing the importance of environmental context in shaping biodiversity–function relationships in decomposer systems (García-Palacios et al. 2011).

This context-dependency was further evident in the contrasting patterns of non-additive effects observed between soils from the two forest types. In NSF soils, which typically contain higher levels of organic matter and support more diverse microbial communities (Yang and Zhu 2015), non-additive effects are more frequent and primarily synergistic, particularly for C mineralization. These results are consistent with those of previous studies demonstrating that functionally diverse litter inputs can enhance microbial activity and nutrient cycling through mechanisms such as nutrient transfer, microbial priming, and the alleviation of stoichiometric limitations (Gartner and Cardon 2004). In comparison, LOP soils, which are often characterized by lower nutrient availability and reduced microbial diversity, exhibited weaker and less consistent non-additive effects. Additive effects were more common in these soils, especially for

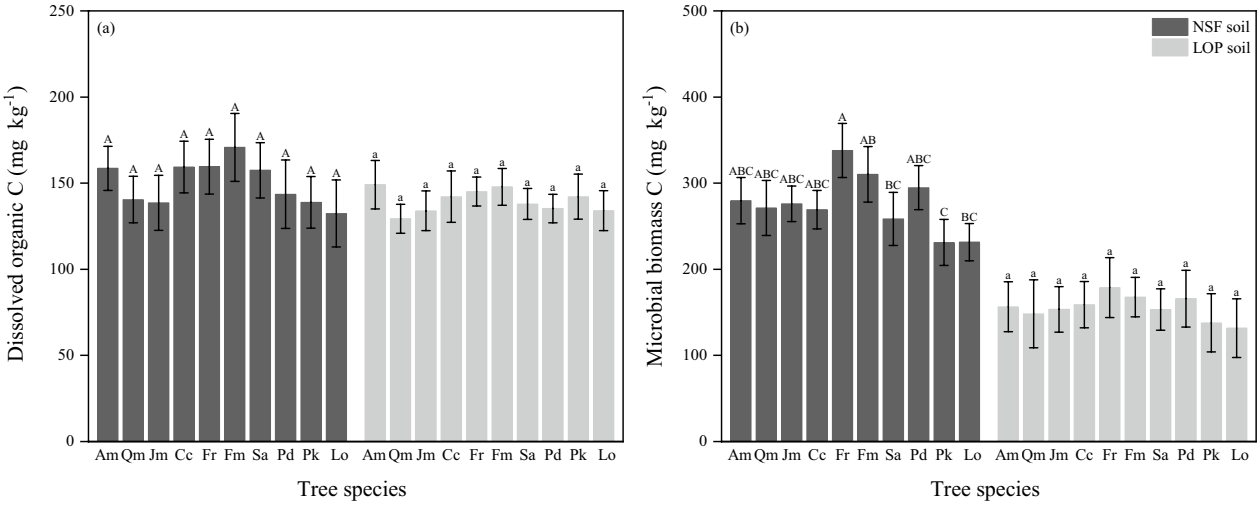


Fig. 2 Effects of different single-species litters on dissolved organic carbon and microbial biomass carbon in natural secondary forests (NSF) and *Larix olgensis* plantations (LOP) soils after 180 days of incubation. (**a**) Dissolved organic carbon (mg kg^{-1}); (**b**) Microbial biomass carbon (mg kg^{-1}). Black bars represent NSF soil, and gray bars represent LOP soil. Different uppercase and lowercase letters above bars indicate significant differences among litter treatments within each soil condition ($P < 0.05$)

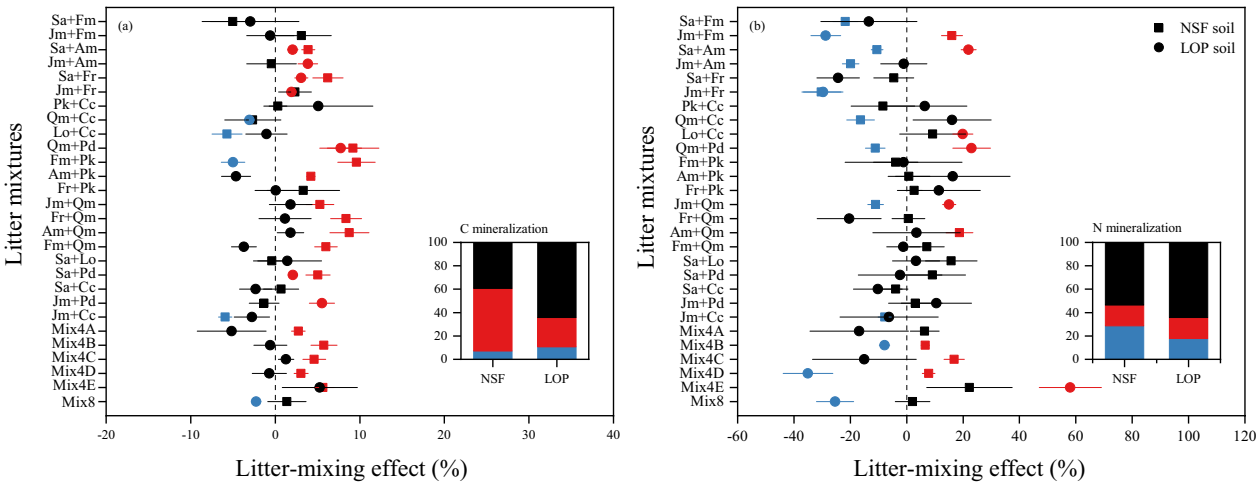


Fig. 3 Litter-mixing effects (%) on soil C and N dynamics in natural secondary forests (NSF) and *Larix olgensis* plantations (LOP) soils after 180 days of incubation. (**a**) carbon (C) mineralization; (**b**) nitrogen (N) mineralization. Litter-mixing effect was calculated as the percentage deviation of the observed value in mixtures from the expected mean of the individual litter effects. Positive values indicate synergistic effects, while negative values indicate antagonistic effects. Squares represent NSF soil, and Circles represent LOP soil. Colored points (red = significant positive effect, blue = significant negative effect) indicate statistically significant mixing effects ($P < 0.05$). Insets show the proportion of mixtures with positive (red), negative (blue), and non-significant (black) mixing effects for each soil conditions. Litter mixture codes refer to specific two-species or four-species combinations, with Mix8 representing eight species

indicators such as MBC. This likely reflects the limited potential of trait-based interactions under more constrained abiotic and biotic conditions (Chen et al. 2025). Together, these results demonstrate that different litter mixtures exert non-additive effects on soil C and N dynamics in different soils.

Relative contributions of functional identity and functional diversity to soil C and N dynamics

A central question in trait-based decomposition ecology is the relative contributions of functional identity (CWM) and functional diversity (Rao's *Q*) to non-additive litter effects (Mokany et al. 2008; Man et al. 2020). Our

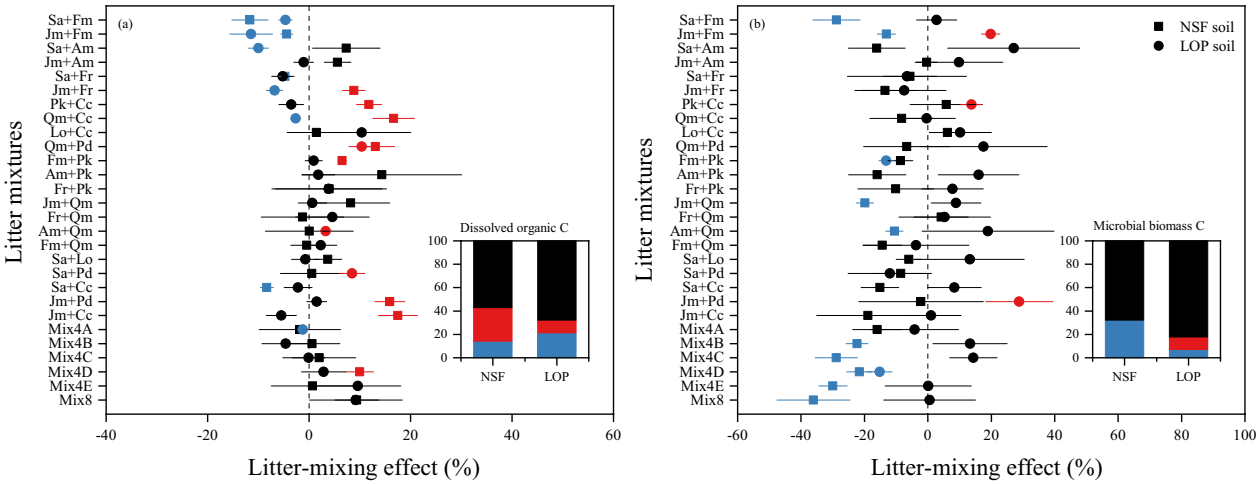


Fig. 4 Litter-mixing effects (%) on dissolved organic carbon and microbial biomass carbon in natural secondary forests (NSF) and *Larix olgensis* plantations (LOP) soils after 180 days of incubation. (**a**) Dissolved organic carbon (DOC); (**b**) Microbial biomass carbon (MBC). Positive values indicate synergistic effects, while negative values indicate antagonistic effects. Squares represent NSF soil, and Circles represent LOP soil. Colored points (red = significant positive effect, blue = significant negative effect) indicate statistically significant mixing effects ($P < 0.05$). Insets show the proportion of mixtures with positive (red), negative (blue), and non-significant (black) mixing effects for each soil conditions. Litter mixture codes refer to specific two-species or four-species combinations, with Mix8 representing eight species

findings indicate that functional diversity is the dominant predictor of non-additive effects in NSF soils, accounting for 89% and 84% of the explained variations in non-additive C and N mineralization, respectively. These findings are consistent with global patterns indicating that higher trait diversity in litter mixtures accelerates decomposition (Kou et al. 2020; Xiao et al. 2020), thereby increasing the input of organic matter into the soil and potentially enhancing nutrient mineralization through improved substrate availability and microbial activity (Wang et al. 2014). The stronger role of functional diversity in NSF soils than LOP soils may be attributed to their higher nutrient status, which provides a wider range of resources to decomposer communities. High nutrient availability alleviates stoichiometric constraints, allowing microbes to exploit complementary litter traits more efficiently (Craine et al. 2007; Mooshammer et al. 2014). This promotes niche differentiation among microbial taxa, with different groups specializing in decomposing distinct litter components (e.g., labile carbohydrates vs. recalcitrant lignin), thereby amplifying the benefits of trait complementarity (Strickland et al. 2009). Additionally, the relatively neutral pH in NSF soils creates favorable conditions for a more diverse and functionally redundant microbial community (Lauber et al. 2009). Neutral to slightly acidic soils generally harbor higher bacterial and fungal diversity, which may broaden the pool of metabolic strategies available for litter decomposition. This diversity increases the likelihood that microbial taxa can match and respond to trait variability among litter species, thereby enhancing non-additive interactions (Hättenschwiler et al. 2005). In contrast, the more acidic conditions of LOP soils can suppress bacterial growth, select for acid-tolerant fungal taxa, and reduce enzymatic efficiency (Rousk et al. 2010), thus limiting the capacity of the microbial community to make better use of trait diversity.

In contrast, trait predictors in LOP soils were more function-specific: functional diversity dominated C mineralization, whereas functional identity played a greater role in N mineralization. This shift likely reflects the limited functional capacity and lower metabolic flexibility of microbial communities in low-fertility soils (Manzoni et al. 2010), which constrain their ability to benefit from trait diversity. Consequently, functional identity becomes more influential, providing consistent and accessible substrates that better match the microbial processing capacity under nutrient constraints. This pattern is consistent with field observations in which functional identity plays a central role in regulating both litter decomposition rates and soil nitrification processes (Laughlin 2011; Lin and Zeng 2018). These findings highlight the roles of functional identity and functional diversity in regulating non-additive effects, with their relative influences shaped by the specific conditions of different forest soils.

Trait-level responses revealed contrasting effects of C-related metrics on non-additive C mineralization. In both soils, Rao's Q of TC exerted a consistent negative effect, suggesting that high interspecific variability in litter C content may hinder microbial decomposition by creating stoichiometric imbalances or

chemically incompatible substrate environments (Vivanco and Austin 2006). High carbon heterogeneity may introduce recalcitrant compounds such as waxes, lignin, or tannins that resist microbial degradation and interfere with the decomposition of more labile components (Palm et al. 2001). Moreover, the positive effects of CWM of cellulose in NSF soils and CWM of hemicellulose in LOP soils highlight the role of structurally distinct yet labile carbon fractions in promoting microbial mineralization. Although cellulose may decompose more efficiently in nutrient-rich soils with diverse microbial communities, hemicellulose, which is more readily hydrolyzed, may be particularly important under resource-limited conditions (Bonanomi et al. 2017; Krishna and Mohan 2017). Rao's *Q* of TN was positively associated with non-additive C mineralization in NSF soils, which is consistent with field-based evidence showing that higher interspecific variability in leaf N content enhances ecosystem respiration and carbon flux (Milcu et al. 2014).

Moreover, Rao's *Q* of TN was positively associated with non-additive N mineralization in soils from both forest types. This suggests that interspecific variability in litter nitrogen content facilitates microbial niche differentiation or resource complementarity during N acquisition. Supporting this, a global meta-analysis by Xiao et al. (2020) found that the effects of litter mixtures on decomposition rates were closely linked to the functional dispersion of N content (ranging from 0 to 2.5) within the litter, underscoring the role of N heterogeneity in driving synergistic decomposition dynamics. In contrast, in LOP soils, CWM of TC and TN were negatively associated with N mineralization, potentially due to microbial immobilization or suppressed mineralization under high C and N, consistent with the findings of Meier and Bowman (2010). However, the mechanisms of C and N mineralization related to microbial communities or enzyme activities rely on indirect evidence from our team's previous studies (Zhang et al. 2019). Future research incorporating microbial community analysis, enzyme assays, and soil biochemical profiling will provide stronger evidence for the microbial processes involved in litter decomposition. Such studies will not only strengthen the robustness of our findings, but also enhance our understanding of how soil and litter characteristics interact in diverse environments.

Practical implications for forest management and biodiversity conservation

Our findings have important implications for forest management and biodiversity conservation. First, the stronger and more frequent non-additive effects observed in natural secondary forest soils underscore the role of tree species diversity in enhancing soil C sequestration and nutrient cycling, suggesting that conserving species-rich broadleaf forests is critical for sustaining ecosystem functions. Second, in nutrient-poor reforested plantations, our results showed that functional identity, rather than functional diversity, played a dominant role in regulating decomposition and nutrient release. This indicates that incorporating tree species with complementary litter traits (e.g., high N or labile C fractions) into reforestation programs can improve soil fertility and accelerate ecosystem recovery in temperate forest ecosystems.

Conclusions

This study demonstrates that the non-additive effects of leaf litter mixtures on soil C and N dynamics are highly context-dependent and shaped by both litter functional traits and different forest soil conditions. Non-additive responses were more frequent and synergistic in NSF soils, but weaker and more variable in LOP soils. Trait-based analyses revealed that functional diversity was the dominant driver of non-additive C and N mineralization in NSF soils, whereas both functional diversity and functional identity played a greater role in LOP soils. Specific traits, such as Rao's *Q* of TN and TC, and CWM of TC, have emerged as key predictors of soil C and N dynamics. These findings highlight the need to integrate both functional identity and diversity within environmentally explicit frameworks to better predict biodiversity–function relationships in decomposer-driven processes. Furthermore, incorporating trait–soil interactions into ecological models may improve our ability to predict nutrient cycling dynamics under shifting biodiversity. From a forest management perspective, our findings suggest that maintaining or enhancing litter functional diversity in secondary forests may aid in sustaining soil fertility, whereas in nutrient-poor plantations, selecting suitable tree species with key functional traits could be more effective in improving nutrient cycling and restoring degraded forest soils.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s13717-025-00658-1.

Supplementary file 1.

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Author contributions
Conceptualization: YK, ZQ; Funding acquisition: YK; Project administration: YK, ZJJ; Methodology: YK, ZQ; Investigation: YK, ZQ, SYJ, ZCJ; Formal analysis: ZQ, SYJ; Supervision: YK, ZJJ; Visualization: ZQ, SYJ; Writing—original draft: ZQ; Writing—review & editing: YK, ZJJ.

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Data availability
The data sets used and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare that they have no competing interests.

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